山东建筑大学 计算机学院 实验报告

班级：软件 212　姓名：孙帅康　学　号：202111110377　实验成绩：_____

课程：计算机网络　同组者：_____　实验日期：2023/5/19

实验三　TCP 和 UDP 协议分析

一、实验目的及任务

1、 熟悉 TCP、UDP 的基本原理
2、 利用 Wireshark 对 TCP、UDP 进行协议分析

二、实验性质及学时

验证性实验，2 学时

三、实验环境

与 Internet 连接的计算机网络系统；操作系统为 windows；Wireshark、IE 等应用软件。

四、实验报告内容（实验题目及答案，其中答案部分标红）

1. 向服务器 gaia.cs.umass.edu 传送文件的客户主机的 IP 地址和 TCP 端口号分别是多少？

515	2023-06-02 09:50:09.573460	10.210.161.235	128.119.245.12	HTTP	1265	POST /wireshark-labs/lab3-1-reply.htm HTTP/1.1 (text/plain)

```
> Frame 515: 1265 bytes on wire (10120 bits), 1265 bytes captured (10120 bits) on interface \Device\NPF_{D78521FB-E3B4
> Ethernet II, Src: IntelCor_e9:79:7e (04:56:e5:e9:79:7e), Dst: RuijieNe_5f:50:9e (58:69:6c:5f:50:9e)
> Internet Protocol Version 4, Src: 10.210.161.235, Dst: 128.119.245.12
> Transmission Control Protocol, Src Port: 54248, Dst Port: 80, Seq: 151870, Ack: 1, Len: 1199
> [127 Reassembled TCP Segments (153068 bytes): #320(747), #321(1222), #322(1222), #323(1222), #324(1222), #325(1222),
v Hypertext Transfer Protocol
  > POST /wireshark-labs/lab3-1-reply.htm HTTP/1.1\r\n
    Host: gaia.cs.umass.edu\r\n
    Connection: keep-alive\r\n
  > Content-Length: 152321\r\n
    Cache-Control: max-age=0\r\n
    Upgrade-Insecure-Requests: 1\r\n
    Origin: http://gaia.cs.umass.edu\r\n
    Content-Type: multipart/form-data; boundary=----WebKitFormBoundaryufmL2eQkiau3GAJD\r\n
    User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/113.0.0.0 Safa
    Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/webp,image/apng,*/*;q=0.8,application/signed-e
    Referer: http://gaia.cs.umass.edu/wireshark-labs/TCP-wireshark-file1.html\r\n
    Accept-Encoding: gzip, deflate\r\n
    Accept-Language: zh-CN,zh;q=0.9,en;q=0.8,en-GB;q=0.7,en-US;q=0.6\r\n
    \r\n
    [Full request URI: http://gaia.cs.umass.edu/wireshark-labs/lab3-1-reply.htm]
    [HTTP request 1/1]
    [Response in frame: 544]
    File Data: 152321 bytes
> MIME Multipart Media Encapsulation, Type: multipart/form-data, Boundary: "----WebKitFormBoundaryufmL2eQkiau3GAJD"
```

答：客户主机 IP：10.210.161.235　　TCP 端口号：54248

2. 服务器 gaia.cs.umass.edu 的 IP 地址是多少？对这一连接，它用来发送和接收 TCP 报文的端口号是多少？

```
> Frame 515: 1265 bytes on wire (10120 bits), 1265 bytes captured (10120 bits) on interface \Device\NPF_{D78521FB-E3B4
> Ethernet II, Src: IntelCor_e9:79:7e (04:56:e5:e9:79:7e), Dst: RuijieNe_5f:50:9e (58:69:6c:5f:50:9e)
> Internet Protocol Version 4, Src: 10.210.161.235, Dst: 128.119.245.12
> Transmission Control Protocol, Src Port: 54248, Dst Port: 80, Seq: 151870, Ack: 1, Len: 1199
> [127 Reassembled TCP Segments (153068 bytes): #320(747), #321(1222), #322(1222), #323(1222), #324(1222), #325(1222),
v Hypertext Transfer Protocol
  > POST /wireshark-labs/lab3-1-reply.htm HTTP/1.1\r\n
    Host: gaia.cs.umass.edu\r\n
    Connection: keep-alive\r\n
```

答：gaia.cs.umass.edu 的 IP 地址：128.119.245.12　TCP 端口号为 80

3. 客户服务器之间用于初始化 TCP 连接的 TCP SYN 报文的序号(sequence number)是多少？在该报文段中，用什么来表示该报文是 SYN 报文段的？

294	2023-06-02 09:50:07.691561	10.210.161.235	39.156.66.18	TCP	74 54293 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM TSval=596591 TSecr=0
295	2023-06-02 09:50:07.716954	39.156.66.18	10.210.161.235	TCP	74 80 → 54293 [SYN, ACK] Seq=0 Ack=1 Win=8192 Len=0 MSS=1234 WS=32 SACK_PERM
296	2023-06-02 09:50:07.717010	10.210.161.235	39.156.66.18	TCP	54 54293 → 80 [ACK] Seq=1 Ack=1 Win=131840 Len=0

答：用于初始化 TCP 连接的 TCP SYN 报文的序号为 0，

```
> Frame 294: 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Device\NPF_{D78521FB-E3B4-4BE3-
> Ethernet II, Src: IntelCor_e9:79:7e (04:56:e5:e9:79:7e), Dst: RuijieNe_5f:50:9e (58:69:6c:5f:50:9e)
> Internet Protocol Version 4, Src: 10.210.161.235, Dst: 39.156.66.18
v Transmission Control Protocol, Src Port: 54293, Dst Port: 80, Seq: 0, Len: 0
    Source Port: 54293
    Destination Port: 80
    [Stream index: 27]
    [Conversation completeness: Complete, NO_DATA (23)]
    [TCP Segment Len: 0]
    Sequence Number: 0    (relative sequence number)
    Sequence Number (raw): 3018652709
    [Next Sequence Number: 1    (relative sequence number)]
    Acknowledgment Number: 0
    Acknowledgment number (raw): 0
    1010 .... = Header Length: 40 bytes (10)
  > Flags: 0x002 (SYN)
    Window: 64240
```

```
Flags: 0x002 (SYN)
    000. .... .... = Reserved: Not set
    ...0 .... .... = Accurate ECN: Not set
    .... 0... .... = Congestion Window Reduced: Not set
    .... .0.. .... = ECN-Echo: Not set
    .... ..0. .... = Urgent: Not set
    .... ...0 .... = Acknowledgment: Not set
    .... .... 0... = Push: Not set
    .... .... .0.. = Reset: Not set
  > .... .... ..1. = Syn: Set
    .... .... ...0 = Fin: Not set
    [TCP Flags: ··********S·]
  Window: 64240
```

用标志字段 SYN=1 来表示。

4. 服务器向客户端发送的 SYNACK 报文段的序号是多少？该报文段中，Acknowledgement Number 字段的值是多少？服务器 gaia.cs.umass.edu 是如何决定此值的？在该报文段中，用什么来标识该报文段是 SYNACK 报文段的？

```
294 2023-06-02 09:50:07.691561 10.210.161.235   39.156.66.18    TCP   74 54293 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM TSval=596591 TSecr=0
295 2023-06-02 09:50:07.716954 39.156.66.18     10.210.161.235  TCP   74 80 → 54293 [SYN, ACK] Seq=0 Ack=1 Win=8192 Len=0 MSS=1234 WS=32 SACK_PERM
296 2023-06-02 09:50:07.717010 10.210.161.235   39.156.66.18    TCP   54 54293 → 80 [ACK] Seq=1 Ack=1 Win=131840 Len=0
```

```
> Ethernet II, Src: RuijieNe_5f:50:9e (58:69:6c:5f:50:9e), Dst: IntelCor_e9:79:7e (04:56:e5:
> Internet Protocol Version 4, Src: 39.156.66.18, Dst: 10.210.161.235
v Transmission Control Protocol, Src Port: 80, Dst Port: 54293, Seq: 0, Ack: 1, Len: 0
    Source Port: 80
    Destination Port: 54293
    [Stream index: 27]
    [Conversation completeness: Complete, NO_DATA (23)]
    [TCP Segment Len: 0]
    Sequence Number: 0    (relative sequence number)
    Sequence Number (raw): 1749922780
    [Next Sequence Number: 1    (relative sequence number)]
    Acknowledgment Number: 1    (relative ack number)
    Acknowledgment number (raw): 3018652710
    1010 .... = Header Length: 40 bytes (10)
  > Flags: 0x012 (SYN, ACK)
    Window: 8192
    [Calculated window size: 8192]
```

答：序号为 0，Acknowledgement Number：1，ACK 字段用于表示确认字段中的值是有效的，说明服务器成功接收了我们发出的连接请求，Acknowledgement 字段的值是 SYN 报文段的 sequence number 加一而来。

```
  1010 .... = Header Length: 40 bytes (10)
  Flags: 0x012 (SYN, ACK)
    000. .... .... = Reserved: Not set
    ...0 .... .... = Accurate ECN: Not set
    .... 0... .... = Congestion Window Reduced: Not set
    .... .0.. .... = ECN-Echo: Not set
    .... ..0. .... = Urgent: Not set
    .... ...1 .... = Acknowledgment: Set
    .... .... 0... = Push: Not set
    .... .... .0.. = Reset: Not set
  > .... .... ..1. = Syn: Set
    .... .... ...0 = Fin: Not set
    [TCP Flags: ·******A··S·]
  Window: 8192
```

5. 包含 HTTP POST 消息请求行的 TCP 报文段的序号是多少？

```
320 2023-06-02 09:50:08.529879 10.210.161.235   128.119.245.12   TCP   813 54248 → 80 [PSH, ACK] Seq=1 Ack=1 Win=515 Len=747 TSval=597429 TSecr=1982423070 [TCP segment o
```

```
> Frame 320: 813 bytes on wire (6504 bits), 813 bytes captured (6504 bits) on interface \Device\NPF_{D78521FB-E3B4-4
> Ethernet II, Src: IntelCor_e9:79:7e (04:56:e5:e9:79:7e), Dst: RuijieNe_5f:50:9e (58:69:6c:5f:50:9e)
> Internet Protocol Version 4, Src: 10.210.161.235, Dst: 128.119.245.12
v Transmission Control Protocol, Src Port: 54248, Dst Port: 80, Seq: 1, Ack: 1, Len: 747
    Source Port: 54248
    Destination Port: 80
    [Stream index: 1]
    [Conversation completeness: Incomplete (30)]
    [TCP Segment Len: 747]
    Sequence Number: 1    (relative sequence number)
    Sequence Number (raw): 297591757
    [Next Sequence Number: 748    (relative sequence number)]
    Acknowledgment Number: 1    (relative ack number)
    Acknowledgment number (raw): 4260211491
    1000 .... = Header Length: 32 bytes (8)
  > Flags: 0x018 (PSH, ACK)
    Window: 515
    [Calculated window size: 515]
    [Window size scaling factor: -1 (unknown)]
    Checksum: 0x2553 [unverified]
```

答：序号为 1

6. 如果将包含 HTTP POST 消息请求行的 TCP 报文段看作是该 TCP 连接从客户到服务器方向上的第一个报文段，那么客户发送给服务器的第六个报文段的序号是多少(从客户到服务器方向)？是何时发送的？服务器对该报文段的 ACK 何时被客户接收的？

```
325 2023-06-02 09:50:08.530132 10.210.161.235   128.119.245.12   TCP   1288 54248 → 80 [ACK] Seq=5636 Ack=1 Win=515 Len=1222 TSval=597429 TSecr=1982423070 [TCP segment of
```

```
> Frame 325: 1288 bytes on wire (10304 bits), 1288 bytes captured (10304 bits) on interface \Device\NPF_{D78521FB-E3
> Ethernet II, Src: IntelCor_e9:79:7e (04:56:e5:e9:79:7e), Dst: RuijieNe_5f:50:9e (58:69:6c:5f:50:9e)
> Internet Protocol Version 4, Src: 10.210.161.235, Dst: 128.119.245.12
v Transmission Control Protocol, Src Port: 54248, Dst Port: 80, Seq: 5636, Ack: 1, Len: 1222
    Source Port: 54248
    Destination Port: 80
    [Stream index: 1]
    [Conversation completeness: Incomplete (30)]
    [TCP Segment Len: 1222]
    Sequence Number: 5636    (relative sequence number)
    Sequence Number (raw): 297597392
    [Next Sequence Number: 6858    (relative sequence number)]
    Acknowledgment Number: 1    (relative ack number)
    Acknowledgment number (raw): 4260211491
    1000 .... = Header Length: 32 bytes (8)
  > Flags: 0x010 (ACK)
    Window: 515
```

答：序号：5636，发送时间：2023-06-02 09:50:08.530132

```
356 2023-06-02 09:50:08.817562 128.119.245.12   10.210.161.235   TCP   66 80 → 54248 [ACK] Seq=1 Ack=748 Win=30464 Len=0 TSval=1982428418 TSecr=597429
357 2023-06-02 09:50:08.817562 128.119.245.12   10.210.161.235   TCP   66 80 → 54248 [ACK] Seq=1 Ack=4414 Win=37888 Len=0 TSval=1982428419 TSecr=597429
358 2023-06-02 09:50:08.817562 128.119.245.12   10.210.161.235   TCP   66 80 → 54248 [ACK] Seq=1 Ack=10524 Win=50048 Len=0 TSval=1982428419 TSecr=597429
359 2023-06-02 09:50:08.817562 128.119.245.12   10.210.161.235   TCP   66 80 → 54248 [ACK] Seq=1 Ack=11746 Win=52992 Len=0 TSval=1982428419 TSecr=597429
```

7. 前六个 TCP 报文段中数据长度各是多少字节？

```
320 2023-06-02 09:50:08.529879 10.210.161.235   128.119.245.12   TCP   813 54248 → 80 [PSH, ACK] Seq=1 Ack=1 Win=515 Len=747 TSval=597429 TSecr=1982423070 [TCP segment o
321 2023-06-02 09:50:08.530132 10.210.161.235   128.119.245.12   TCP   1288 54248 → 80 [ACK] Seq=748 Ack=1 Win=515 Len=1222 TSval=597429 TSecr=1982423070 [TCP segment of
322 2023-06-02 09:50:08.530132 10.210.161.235   128.119.245.12   TCP   1288 54248 → 80 [ACK] Seq=1970 Ack=1 Win=515 Len=1222 TSval=597429 TSecr=1982423070 [TCP segment of
323 2023-06-02 09:50:08.530132 10.210.161.235   128.119.245.12   TCP   1288 54248 → 80 [ACK] Seq=3192 Ack=1 Win=515 Len=1222 TSval=597429 TSecr=1982423070 [TCP segment of
324 2023-06-02 09:50:08.530132 10.210.161.235   128.119.245.12   TCP   1288 54248 → 80 [ACK] Seq=4414 Ack=1 Win=515 Len=1222 TSval=597429 TSecr=1982423070 [TCP segment of
325 2023-06-02 09:50:08.530132 10.210.161.235   128.119.245.12   TCP   1288 54248 → 80 [ACK] Seq=5636 Ack=1 Win=515 Len=1222 TSval=597429 TSecr=1982423070 [TCP segment of
```

答：1222

8. 在整个跟踪过程中，服务器向客户机通知的最小可用缓存是多少？在整个跟踪过程中是否发生过由于服务器可用缓存(接收方缓存)不够用而限制发送方的发送速率的情况？

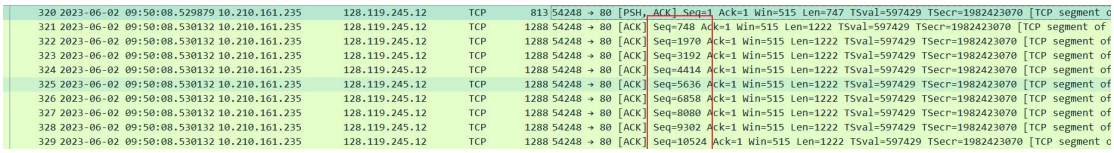
```
356 2023-06-02 09:50:08.817562 128.119.245.12   10.210.161.235   TCP   66 80 → 54248 [ACK] Seq=1 Ack=748 Win=30464 Len=0 TSval=1982428418 TSecr=597429
```

答：30464

没有发生

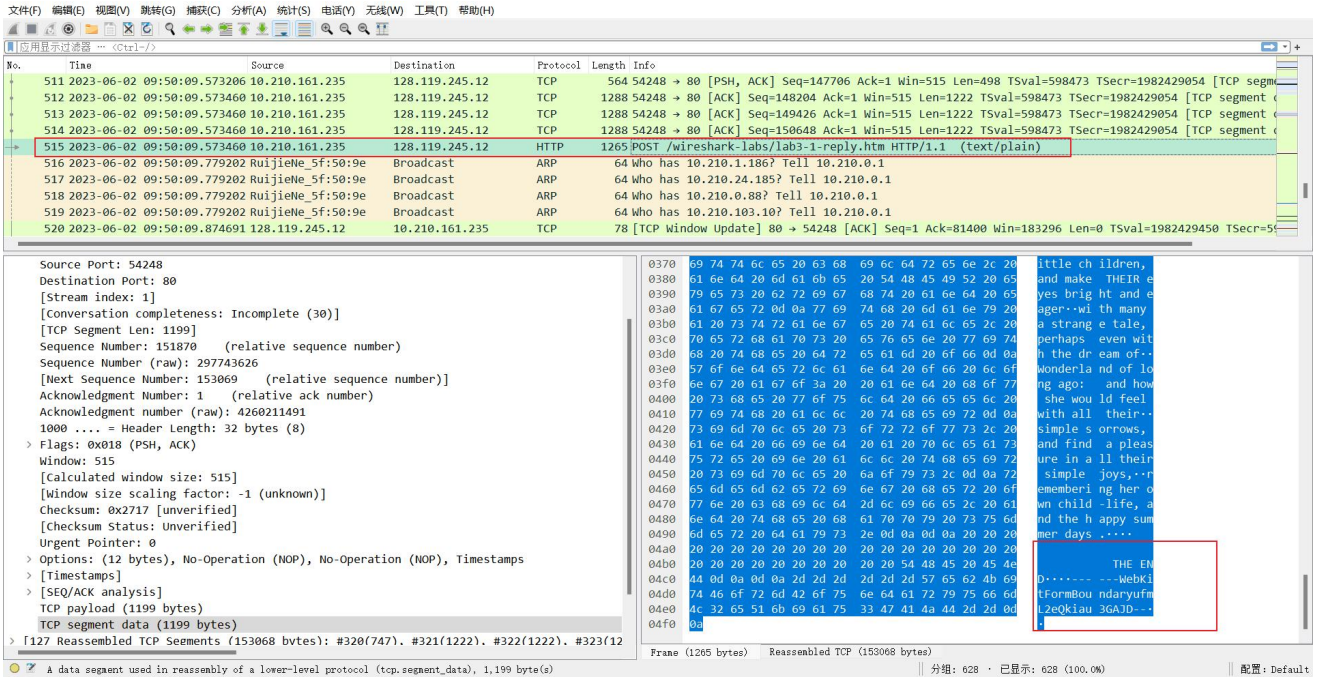
9. 在跟踪文件中是否有重传的报文段？判断的依据是什么？

答：没有，从 TCP 报文段的序列号中可以看出。从源端发往目的端的序号逐渐递增，如果这其中有重传的报文段，则其序号中应该有小于其临近的分组序号的分组，在图中未看到这样的分组，所以没有被重传的分组。



320	2023-06-02 09:50:08.529879	10.210.161.235	128.119.245.12	TCP	813	54248 → 80	[PSH, ACK] Seq=1 Ack=1 Win=515 Len=747 TSval=597429 TSecr=1982423070 [TCP segment o
321	2023-06-02 09:50:08.530132	10.210.161.235	128.119.245.12	TCP	1288	54248 → 80	[ACK] Seq=748 Ack=1 Win=515 Len=1222 TSval=597429 TSecr=1982423070 [TCP segment of
322	2023-06-02 09:50:08.530132	10.210.161.235	128.119.245.12	TCP	1288	54248 → 80	[ACK] Seq=1970 Ack=1 Win=515 Len=1222 TSval=597429 TSecr=1982423070 [TCP segment of
323	2023-06-02 09:50:08.530132	10.210.161.235	128.119.245.12	TCP	1288	54248 → 80	[ACK] Seq=3192 Ack=1 Win=515 Len=1222 TSval=597429 TSecr=1982423070 [TCP segment of
324	2023-06-02 09:50:08.530132	10.210.161.235	128.119.245.12	TCP	1288	54248 → 80	[ACK] Seq=4414 Ack=1 Win=515 Len=1222 TSval=597429 TSecr=1982423070 [TCP segment of
325	2023-06-02 09:50:08.530132	10.210.161.235	128.119.245.12	TCP	1288	54248 → 80	[ACK] Seq=5636 Ack=1 Win=515 Len=1222 TSval=597429 TSecr=1982423070 [TCP segment of
326	2023-06-02 09:50:08.530132	10.210.161.235	128.119.245.12	TCP	1288	54248 → 80	[ACK] Seq=6858 Ack=1 Win=515 Len=1222 TSval=597429 TSecr=1982423070 [TCP segment of
327	2023-06-02 09:50:08.530132	10.210.161.235	128.119.245.12	TCP	1288	54248 → 80	[ACK] Seq=8080 Ack=1 Win=515 Len=1222 TSval=597429 TSecr=1982423070 [TCP segment of
328	2023-06-02 09:50:08.530132	10.210.161.235	128.119.245.12	TCP	1288	54248 → 80	[ACK] Seq=9302 Ack=1 Win=515 Len=1222 TSval=597429 TSecr=1982423070 [TCP segment of
329	2023-06-02 09:50:08.530132	10.210.161.235	128.119.245.12	TCP	1288	54248 → 80	[ACK] Seq=10524 Ack=1 Win=515 Len=1222 TSval=597429 TSecr=1982423070 [TCP segment o

10. TCP 连接的吞吐率(bytes transferred per unit time，单位时间传输的字节数)是多少？写出计算方法和计算过程。



No.	Time	Source	Destination	Protocol	Length	Info
511	2023-06-02 09:50:09.573206	10.210.161.235	128.119.245.12	TCP	564	54248 → 80 [PSH, ACK] Seq=147706 Ack=1 Win=515 Len=498 TSval=598473 TSecr=1982429054 [TCP segm
512	2023-06-02 09:50:09.573460	10.210.161.235	128.119.245.12	TCP	1288	54248 → 80 [ACK] Seq=148204 Ack=1 Win=515 Len=1222 TSval=598473 TSecr=1982429054 [TCP segment
513	2023-06-02 09:50:09.573460	10.210.161.235	128.119.245.12	TCP	1288	54248 → 80 [ACK] Seq=149426 Ack=1 Win=515 Len=1222 TSval=598473 TSecr=1982429054 [TCP segment
514	2023-06-02 09:50:09.573460	10.210.161.235	128.119.245.12	TCP	1288	54248 → 80 [ACK] Seq=150648 Ack=1 Win=515 Len=1222 TSval=598473 TSecr=1982429054 [TCP segment
515	2023-06-02 09:50:09.573460	10.210.161.235	128.119.245.12	HTTP	1265	POST /wireshark-labs/lab3-1-reply.htm HTTP/1.1 (text/plain)
516	2023-06-02 09:50:09.779202	RuijieNe_5f:50:9e	Broadcast	ARP	64	Who has 10.210.1.186? Tell 10.210.0.1
517	2023-06-02 09:50:09.779202	RuijieNe_5f:50:9e	Broadcast	ARP	64	Who has 10.210.24.185? Tell 10.210.0.1
518	2023-06-02 09:50:09.779202	RuijieNe_5f:50:9e	Broadcast	ARP	64	Who has 10.210.0.88? Tell 10.210.0.1
519	2023-06-02 09:50:09.779202	RuijieNe_5f:50:9e	Broadcast	ARP	64	Who has 10.210.103.10? Tell 10.210.0.1
520	2023-06-02 09:50:09.874691	128.119.245.12	10.210.161.235	TCP	78	[TCP Window Update] 80 → 54248 [ACK] Seq=1 Ack=81400 Win=183296 Len=0 TSval=1982429450 TSecr=5

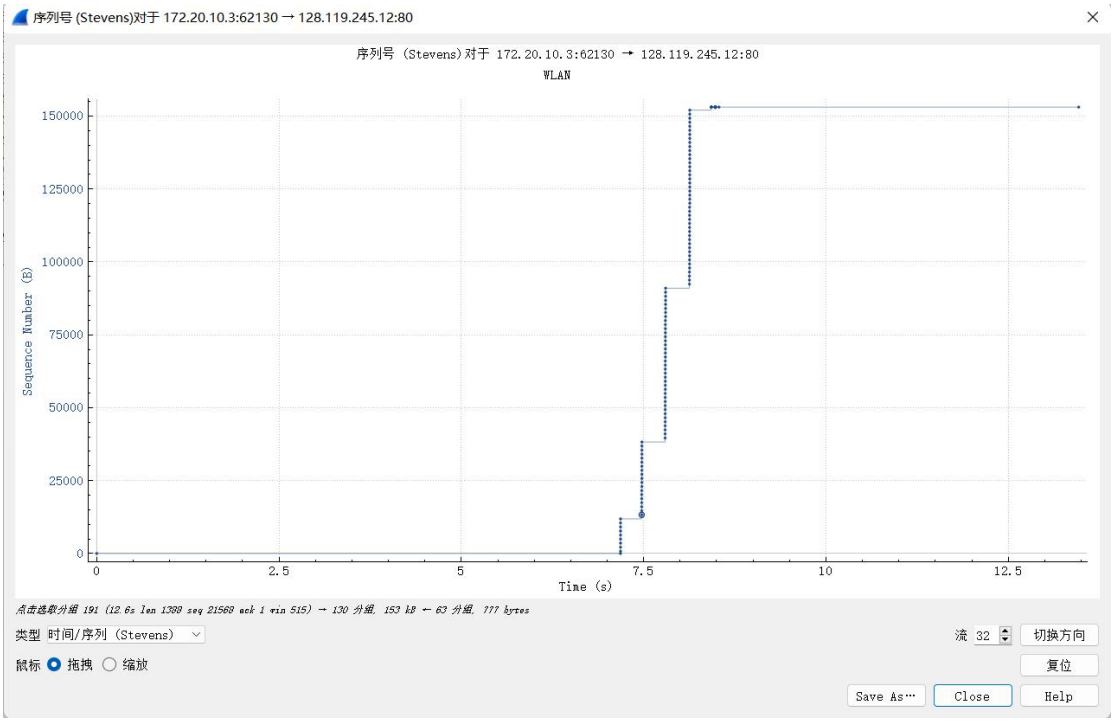
```
Source Port: 54248
Destination Port: 80
[Stream index: 1]
[Conversation completeness: Incomplete (30)]
[TCP Segment Len: 1199]
Sequence Number: 151870    (relative sequence number)
Sequence Number (raw): 297743626
[Next Sequence Number: 153069    (relative sequence number)]
Acknowledgment Number: 1    (relative ack number)
Acknowledgment number (raw): 4260211491
1000 .... = Header Length: 32 bytes (8)
> Flags: 0x018 (PSH, ACK)
Window: 515
[Calculated window size: 515]
[Window size scaling factor: -1 (unknown)]
Checksum: 0x2717 [unverified]
[Checksum Status: Unverified]
Urgent Pointer: 0
> Options: (12 bytes), No-Operation (NOP), No-Operation (NOP), Timestamps
> [Timestamps]
> [SEQ/ACK analysis]
TCP payload (1199 bytes)
TCP segment data (1199 bytes)
> [127 Reassembled TCP Segments (153068 bytes): #320(747), #321(1222), #322(1222), #323(12
```

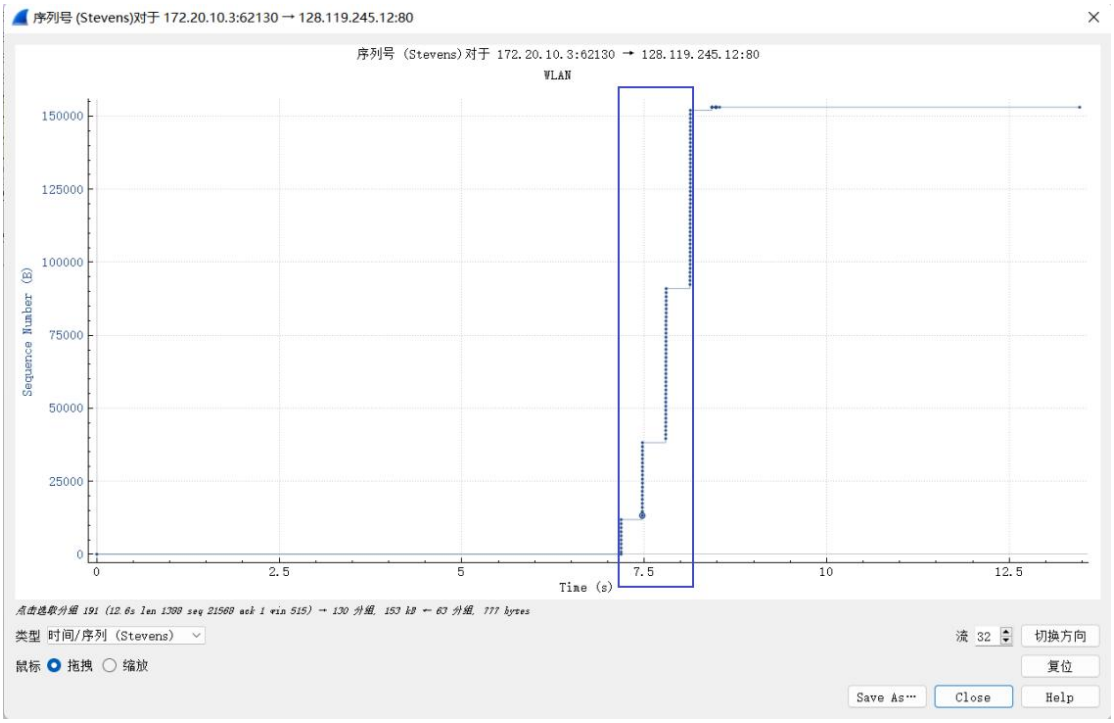
时间间隔：9.573460-8.529879=1.043581s

数据长度：150648B

吞吐率：150648B/1.043581s=144356.79bps

11. 首先，在捕获分组列表子窗口中选择一个从客户发给服务器的报文；然后，利用 Time-Sequence-Graph(Stevens) plotting 工具，浏览由客户端向服务器发送的报文段序号和时间的对应关系图。你能否辨别出 TCP 慢启动阶段的起始和结束时间，以及发送端在图中何处/何时转入避免拥塞阶段的？





答：慢启动 12.31s 到 13.26s，

无法判断拥塞避免阶段。

12. 阐述通过跟踪观察到的 TCP 发送方的行为与理想化的 TCP 发送方的行为有何不同？

答：在慢启动阶段时，如果有拥塞的话，发送方会根据 AIMD 算法进行调整。在此实验中，在发送方还可以发送数据的时候，已经没有数据需要发送了。在 Web 应用中，有些 web 对象比较小，在慢启动还没有结束之前，传送就已经结束了，所以传送小的 web 对象受到 TCP 慢启动阶段的影响，导致较长的延迟。

13. 选择任意一个 UDP 报文段。该 UDP 报文段的头部有哪几个字段？

```
User Datagram Protocol, Src Port: 60221, Dst Port: 53
    Source Port: 60221
    Destination Port: 53
    Length: 42
    Checksum: 0x6c68 [unverified]
    [Checksum Status: Unverified]
    [Stream index: 0]
  > [Timestamps]
    UDP payload (34 bytes)
> Domain Name System (query)
```

答：源端口号：60221，目的端口号：53，长度：42，检验和。

14. 接上题。该 UDP 报文段中每个头部字段的长度为多少字节？

答：头部字段长度=Length- UDP payload=42-34=8Byte

15. 接上题。该 UDP 报文段头部的 length 字段的值是多少？

答：42

16. 接上题。UDP 报文段有效载荷的最大长度可以是多少字节？

答：34

17. 接上题。最大可能的源端口号是多少？

答：60221

18. 接上题。UDP 的协议号是多少？(为回答该问题，需要检查封装 UDP 报文段的 IP 数据报的头部字段 protocol)

```
Internet Protocol Version 4, Src: 172.20.10.3, Dst: 172.20.10.1
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
  > Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    Total Length: 62
    Identification: 0x8f60 (36704)
  > 000. .... = Flags: 0x0
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 128
    Protocol: UDP (17)
    Header Checksum: 0x0000 [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 172.20.10.3
    Destination Address: 172.20.10.1
```

答：UDP（17）

19. 检查一对 UDP 报文段(第二个段是对第一个段的响应)。两个 UDP 报文段中的端口号之间的关系是什么？

```
> Frame 95: 76 bytes on wire (608 bits), 76 bytes captured (608 bits) on interface \Device\NPF_{D78521FB-E3B4-4BE3
> Ethernet II, Src: IntelCor_e9:79:7e (04:56:e5:e9:79:7e), Dst: 9e:fc:28:47:64:64 (9e:fc:28:47:64:64)
> Internet Protocol Version 4, Src: 172.20.10.3, Dst: 172.20.10.1
> User Datagram Protocol, Src Port: 60221, Dst Port: 53
> Domain Name System (query)
```

```
> Frame 97: 171 bytes on wire (1368 bits), 171 bytes captured (1368 bits) on interface \Device\NPF_{D78521FB-E3B4-
> Ethernet II, Src: 9e:fc:28:47:64:64 (9e:fc:28:47:64:64), Dst: IntelCor_e9:79:7e (04:56:e5:e9:79:7e)
> Internet Protocol Version 4, Src: 172.20.10.1, Dst: 172.20.10.3
> User Datagram Protocol, Src Port: 53, Dst Port: 60221
> Domain Name System (response)
```

答：第一个段：源端口号：60221，目的端口号：53，

第二个段：源端口号：53，目的端口号：60221。